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Studierichting: Informatica
Oefeningenreeks 5G nr. 17

$$y^2 = 4x \Rightarrow x = \frac{y^2}{4}$$

snijpunten met $x = 1$:

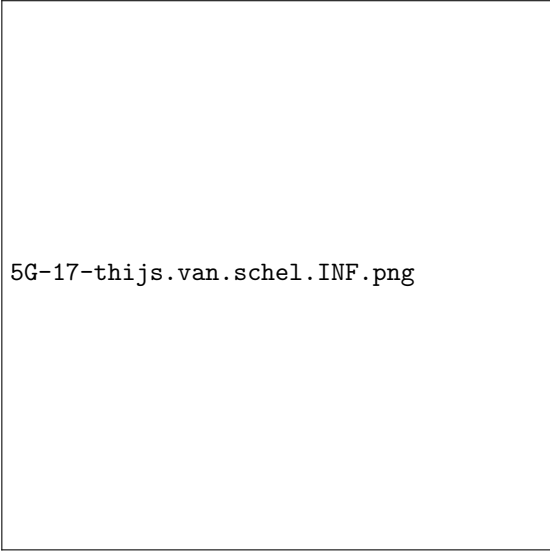
$$y^2 = 4 \cdot 1 \Rightarrow y^2 = 4 \Rightarrow y = -2 \text{ of } y = 2$$

omwenteling rond $x = 1$, integratie naar y :

$$\begin{aligned} V &= \pi \int_{-2}^2 (1-x)^2 dy \\ &= \pi \int_{-2}^2 \left(1 - \frac{y^2}{4}\right)^2 dy \\ &= \pi \int_{-2}^2 \left(1 - \frac{y^2}{2} + \frac{y^4}{16}\right) dy \\ &= \pi \left[y - \frac{y^3}{6} + \frac{y^5}{80} \right]_{-2}^2 \\ &= \pi \left(\left(2 - \frac{8}{6} + \frac{32}{80}\right) - \left(-2 - \frac{-8}{6} + \frac{-32}{80}\right) \right) \\ &= \pi \left(\left(2 - \frac{4}{3} + \frac{2}{5}\right) - \left(-2 + \frac{4}{3} - \frac{2}{5}\right) \right) \\ &= \pi \left(4 - \frac{8}{3} + \frac{4}{5}\right) \\ &= \pi \left(\frac{60}{15} - \frac{40}{15} + \frac{12}{15}\right) \\ V &= \frac{32\pi}{15} \end{aligned}$$

tekening:

References



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